Hospital Management System with Telemedicine –

Detailed Requirement Analysis

Project Overview:

The Hospital Management System (HMS) with Telemedicine is an advanced system to manage hospital operations and facilitate telemedicine for remote healthcare services. The system will manage patients, doctors, medical records, billing, appointments, prescriptions, and remote consultations through video conferencing. It will streamline workflows and improve patient care by integrating multiple hospital departments.

Entities:

1. **Patient:**
    - Attributes:
        - patientId: Unique identifier for the patient.
        - name, age, gender: Basic demographic details.
        - contactInfo: Contact details including phone, email.
        - address: Residential address.
        - medicalHistory: Past medical history of the patient.
        - insuranceDetails: Insurance information.
    - Relationships:
        - A patient can have multiple Appointments, TeleConsultations, Prescriptions, MedicalRecords, and LabReports.
2. **Doctor:**
    - Attributes:
        - doctorId: Unique identifier for each doctor.
        - name, specialization: Basic details.
        - contactInfo: Contact information for doctor.
        - experience: Years of experience.
        - availability: Time slots and availability for appointments.
    - Relationships:
        - A doctor can handle multiple Appointments, Prescriptions, TeleConsultations, and be assigned to multiple Departments.
3. **TeleConsultation:**
    - Attributes:
        - teleConsultationId: Unique identifier for the telemedicine session.
        - patientId, doctorId: Relationship to patient and doctor.
        - appointmentDateTime: Date and time of consultation.

        - videoLink: Video session link.
        - consultationNotes: Notes written by the doctor during/after the session.

4. **Appointment:**
    - Attributes:
        - appointmentId: Unique identifier for each appointment.
        - patientId, doctorId: Link to patient and doctor.
        - appointmentType: Type (In-person, Telemedicine).
        - appointmentDateTime: Scheduled date and time.
        - status: Status (Pending, Completed, Cancelled).
5. **Prescription:**
    - Attributes:
        - prescriptionId: Unique identifier.
        - doctorId, patientId: Linked to the doctor and patient.
        - medicineList: List of medicines prescribed.
        - dateIssued: Date of prescription.
6. **MedicalRecord:**
    - Attributes:
        - recordId: Unique record identifier.
        - patientId: Linked to a patient.
        - medicalHistory, diagnosis, treatmentPlan: Detailed medical information.
7. **LabReport:**
    - Attributes:
        - labReportId: Unique identifier for the report.
        - patientId: Linked to a patient.
        - testName, testResults: Lab test information.
        - reportDate: Date the report was generated.
8. **Payment:**
    - Attributes:
        - paymentId: Unique identifier.
        - amount: Payment amount.
        - paymentDate: Date of payment.
        - paymentType: Method (Credit, Cash, Insurance).
        - invoiceId: Link to the invoice.
9. **Insurance:**
    - Attributes:

        - insuranceId: Unique identifier.
        - patientId: Linked to a patient.
        - insuranceProvider: Name of the insurance company.
        - policyNumber: Policy details.
        - claimStatus: Status of insurance claim.

10. **Pharmacy:**
    - Attributes:
        - pharmacyId: Unique identifier.
        - pharmacyName: Name of the pharmacy.
        - location: Address of the pharmacy.
        - contactInfo: Contact details.
11. **Medicine:**
    - Attributes:
        - medicineId: Unique identifier.
        - name: Name of the medicine.
        - stock: Quantity in stock.
        - price: Price per unit.
12. **Staff:**
    - Attributes:
        - staffId: Unique identifier.
        - name, role, contactInfo: Basic details.
        - shiftSchedule: Work hours and shifts.
        - departmentId: Linked to a department.
13. **Department:**
    - Attributes:
        - departmentId: Unique identifier.
        - departmentName: Name of the department (Cardiology, Neurology, etc.).
        - headOfDepartment: Doctor responsible for the department.
14. **Room:**
    - Attributes:
        - roomId: Unique identifier.
        - roomType: Type of room (General, ICU, Private).
        - availabilityStatus: Available or Occupied.
        - assignedPatientId: Linked to patient when occupied.
15. **Invoice:**

- Attributes:
  - invoiceId: Unique identifier.
  - patientId: Linked to the patient.
  - amountDue: Total amount to be paid.
  - dueDate: Payment due date.

Modules:

1. **Telemedicine:**
   - **Purpose**: Facilitate remote consultations via video conferencing.
   - **Features**:
     - Schedule and conduct video consultations between doctors and patients.
     - Secure video link generation and access.
     - Consultation notes and prescriptions issued during or after the session.
     - Option for patients to provide feedback.
2. **Appointment Scheduling:**
   - **Purpose**: Manage in-person and telemedicine appointments.
   - **Features**:
     - Schedule appointments based on doctor availability.
     - Distinguish between in-person and telemedicine appointments.
     - Appointment reminder notifications via email/SMS.
3. **Medical Records Management:**
   - **Purpose**: Centralize the tracking of patient history, lab reports, and prescriptions.
   - **Features**:
     - Secure storage and retrieval of medical records.
     - Easy access to lab reports and medical history for both patients and doctors.
     - Integration with other modules to update medical records based on new lab results, prescriptions, etc.
4. **Billing and Insurance:**
   - **Purpose**: Handle billing, payments, and insurance claims.
   - **Features**:
     - Generate invoices for medical services provided.
     - Manage payments and track outstanding bills.
     - Handle insurance claims and link payments to insurance providers.
5. **Pharmacy Management:**
   - **Purpose**: Manage hospital pharmacy and track prescriptions.

- **Features**:
    - Maintain stock of medicines and generate alerts when stock is low.
    - Process prescriptions issued by doctors.
    - Track pharmacy sales and manage billing.

6. **Room and Staff Management:**
    - **Purpose**: Manage hospital rooms and schedule staff efficiently.
    - **Features**:
        - Assign rooms to patients based on availability.
        - Track room occupancy and availability status.
        - Manage staff schedules, shift allocations, and department assignments.
7. **Reporting Module:**
    - **Purpose**: Generate detailed reports on various aspects of hospital performance.
    - **Features**:
        - Patient health reports and progress over time.
        - Financial reports including billing and payment analytics.
        - Hospital performance analytics (occupancy rates, appointment success rates, etc.).

Technology Stack:

- **Backend**: Core Java
- **ORM**: Hibernate for database interaction
- **Database**: MySQL
- **Frontend**: (Optional if you want a UI) HTML, CSS, JS
- **Others**: Maven for project dependency management

Detailed module description

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1. Telemedicine Module

Purpose: To facilitate remote consultations between doctors and patients using video calls. This module handles scheduling, conducting, and managing remote consultations.

Features:

- **TeleConsultation Scheduling**: Allow patients to book teleconsultations based on doctors' availability.
- **Video Conferencing Integration**: Enable video sessions for remote doctor-patient interactions.
- **Consultation History**: Maintain records of previous teleconsultations, including doctor notes and outcomes.
- **Prescription Issuance**: Doctors can issue prescriptions digitally during or after the teleconsultation.
- **Payment Integration**: Patients can make payments before or after the teleconsultation.
- **Notifications**: Send reminders to both doctors and patients about upcoming teleconsultations.

Key Data Flows:

- **Appointment** ↔ **TeleConsultation** (One-to-One)
- **Doctor** ↔ **TeleConsultation** (One-to-Many)
- **Patient** ↔ **TeleConsultation** (One-to-Many)

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2. Appointment Scheduling Module

Purpose: To manage both in-person and telemedicine appointments between patients and doctors. This module tracks doctor availability, patient requests, and handles appointment confirmation.

Features:

- **Appointment Booking**: Patients can book appointments for in-person visits or telemedicine consultations.
- **Doctor Availability Management**: Doctors can manage their available slots for both types of appointments.
- **Patient History**: View a patient's previous appointments and link appointments to their medical records.
- **Rescheduling and Cancellation**: Allow patients and doctors to reschedule or cancel appointments.
- **Appointment Reminders**: Automated notifications for upcoming appointments via email or SMS.

Key Data Flows:

- **Patient** ↔ **Appointment** (One-to-Many)
- **Doctor** ↔ **Appointment** (One-to-Many)
- **Appointment** ↔ **TeleConsultation** (One-to-One for telemedicine)

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3. Medical Records Management Module

Purpose: To keep a consolidated record of patients' medical history, including their diagnoses, prescriptions, lab results, and past consultations. This is essential for continuity of care and legal compliance.

Features:

- **Patient Medical Records**: Store and manage detailed medical histories, including past treatments, prescriptions, and diagnoses.
- **Lab Report Management**: Track all lab reports and attach them to the patient's medical record.
- **Prescription History**: Maintain an archive of all prescriptions issued to the patient.
- **Access Control**: Ensure only authorized doctors and staff can access sensitive patient medical information.
- **Medical Reports**: Generate detailed medical reports for doctors or patients when needed.

Key Data Flows:

- **Patient** ↔ **MedicalRecord** (One-to-One)
- **MedicalRecord** ↔ **LabReport** (One-to-Many)
- **MedicalRecord** ↔ **Prescription** (One-to-Many)

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4. Billing and Insurance Module

Purpose: To handle all financial aspects of the hospital, including billing for services rendered (consultations, treatments, prescriptions) and insurance claims processing.

Features:

- **Invoice Generation**: Automatically generate invoices for each consultation, prescription, lab test, and other hospital services.
- **Payment Management**: Track patient payments (both full and partial payments), including integration with online payment gateways.
- **Insurance Claims**: Handle insurance claims submission and tracking, including partial payments by insurance companies.
- **Invoice History**: Maintain a history of all past invoices for each patient.
- **Discount and Refund Processing**: Manage discounts (if applicable) and issue refunds when necessary.

Key Data Flows:

- **Patient** ↔ **Invoice** (One-to-Many)
- **Invoice** ↔ **Payment** (One-to-One)
- **Patient** ↔ **Insurance** (One-to-One)

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5. Pharmacy Management Module

Purpose: To manage the inventory of medicines in the hospital pharmacy, track prescriptions issued by doctors, and handle dispensation to patients.

Features:

- **Medicine Inventory**: Track the stock levels of medicines, with alerts for low-stock or expired items.

- **Prescription Fulfillment**: When a doctor issues a prescription, the system links it to the pharmacy for dispensation.
- **Medicine Request and Delivery**: Patients can request home delivery of medicines for telemedicine consultations.
- **Pharmacy Reports**: Generate reports on the usage and sale of medicines over time.
- **Supplier Management**: Manage suppliers for medicines and reorder stock when necessary.

Key Data Flows:

- **Prescription** ↔ **Pharmacy** (One-to-Many)
- **Pharmacy** ↔ **Medicine** (One-to-Many)

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6. Room and Staff Management Module

Purpose: To manage the assignment of hospital rooms to admitted patients, as well as the scheduling and shift management of hospital staff.

Features:

- **Room Assignment**: Assign patients to available rooms (for admitted cases), track occupied and available rooms.
- **Room Status Management**: Track the status of rooms (vacant, occupied, under maintenance).
- **Staff Scheduling**: Manage shifts for doctors, nurses, and other hospital staff. Ensure there is adequate staffing coverage at all times.
- **Staff Availability**: Track which staff members are available for duty on any given day.
- **Overtime and Leave Tracking**: Track overtime and leave requests for staff.

Key Data Flows:

- **Patient** ↔ **Room** (One-to-One)
- **Department** ↔ **Staff** (One-to-Many)

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7. Reporting and Analytics Module

Purpose: To provide actionable insights and data analysis for hospital administrators, doctors, and other stakeholders through various types of reports, such as patient health, hospital performance, and financial analytics.

Features:

- **Patient Reports**: Generate detailed reports on a patient's health history, including medical records, lab results, and prescriptions.
- **Financial Reports**: Track hospital revenues and expenses, including billing details and payment status.
- **Staff Performance Reports**: Analyze staff performance in terms of patient load, shifts worked, and consultation outcomes.
- **Hospital Utilization Reports**: Monitor room occupancy rates, doctor appointment schedules, and patient admission rates.
- **Medicine Usage Reports**: Track which medicines are being prescribed the most and monitor inventory usage.

Key Data Flows:

- **Patient** ↔ **MedicalRecord**, **LabReport**, **Prescription** (For health-related reports)
- **Invoice** ↔ **Payment** (For financial reports)
- **Staff** ↔ **Department** (For performance reports)

Summary of Data Flows Across Modules:

Module	**Entities Involved**
Telemedicine Module	TeleConsultation, Appointment, Doctor, Patient
Appointment Scheduling	Appointment, Doctor, Patient
Medical Records Management	MedicalRecord, LabReport, Prescription, Patient
Billing and Insurance	Invoice, Payment, Insurance, Patient
Pharmacy Management	Pharmacy, Prescription, Medicine
Room and Staff Management	Room, Staff, Patient, Department
Reporting and Analytics	Patient, MedicalRecord, Prescription, LabReport, Invoice, Payment, Staff, Department

Detailed workflow

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1. Patient Registration & Login Workflow

Steps:

1. **Patient Registration**:
    - The patient registers via the online portal by providing personal details such as name, contact, address, date of birth, insurance details, etc.
    - System stores the data in the Patient table.
    - A unique patientId is generated.
2. **Patient Login**:
    - The patient logs in using their credentials.
    - The system validates the login details and retrieves the patient's profile from the Patient table.

Data Flow:

- Patient → Patient table

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2. Doctor Registration & Login Workflow

Steps:

1. **Doctor Registration**:
    - Doctors are either registered by the hospital administration or self-register.
    - Each doctor provides qualifications, department, contact information, and specialization.
    - Data is stored in the Doctor table, and a unique doctorId is generated.
2. **Doctor Login**:
    - Doctors log in to the system using their credentials.
    - The system retrieves their profile from the Doctor table.

Data Flow:

- Doctor → Doctor table

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3. Appointment Scheduling Workflow

Steps:

1. **Patient Requests an Appointment**:
    - The patient selects either an in-person or telemedicine appointment.
    - They choose a doctor and preferred time slot based on the doctor's availability.
    - An appointment request is created and stored in the Appointment table.
2. **Doctor Reviews Appointment Requests**:

- The doctor reviews their appointment requests.
- The doctor can accept, reject, or reschedule the appointment.

3. **Confirmation**:
   - Once confirmed, the appointment is set, and both the patient and doctor receive notifications.
   - The confirmed appointment is updated in the Appointment table.

Data Flow:

- Patient → Appointment Request → Appointment table
- Doctor → Approves → Appointment table

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4. Telemedicine Workflow

Steps:

1. **TeleConsultation Setup**:
   - For a telemedicine appointment, the system creates an entry in the TeleConsultation table linked to the Appointment.
   - A video conferencing link is generated and sent to both the patient and doctor.
2. **TeleConsultation Session**:
   - At the scheduled time, the doctor and patient join the video session.
   - During the consultation, the doctor can view the patient's medical records and history.
3. **Post-Consultation**:
   - After the session, the doctor can issue prescriptions, request lab tests, or add notes to the medical record.
   - Data is updated in the TeleConsultation, Prescription, and MedicalRecord tables.

Data Flow:

- Appointment → TeleConsultation table
- Doctor → Prescription → Prescription table
- Doctor → Medical Notes → MedicalRecord table

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5. Medical Records Management Workflow

Steps:

1. **Patient Medical Record Access**:
   - Doctors can view the patient's medical history (prescriptions, lab reports, past consultations).
   - The system retrieves all relevant data from the MedicalRecord, LabReport, and Prescription tables.
2. **Updating Medical Records**:
   - After each consultation, the doctor can update the patient's medical record with new diagnoses, prescriptions, or lab results.

- These updates are saved in the MedicalRecord and Prescription tables.

3. **Lab Reports Management**:
    - Lab technicians can upload lab results to the system, which are then linked to the patient's medical record.

Data Flow:

- Doctor → Medical History → MedicalRecord, Prescription, LabReport tables
- Doctor → Update Record → MedicalRecord table
- Lab → Upload Lab Report → LabReport table

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6. Billing and Insurance Workflow

Steps:

1. **Invoice Generation**:
    - After a consultation (in-person or telemedicine) or after treatment, the system generates an invoice for the services provided.
    - The invoice is stored in the Invoice table.
2. **Insurance Claim Submission**:
    - If the patient has insurance, an insurance claim is generated and submitted based on the invoice.
    - The insurance claim is processed and tracked in the Insurance and Invoice tables.
3. **Payment Processing**:
    - The patient or their insurance provider processes the payment via the system's payment gateway.
    - The payment is linked to the invoice, and the status is updated in the Payment table.

Data Flow:

- Consultation → Invoice → Invoice table
- Insurance Claim → Insurance and Invoice tables
- Payment → Payment table

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7. Pharmacy Management Workflow

Steps:

1. **Prescription Generation**:
    - After a consultation, the doctor issues a prescription.
    - The prescription is stored in the Prescription table.
2. **Medicine Dispensation**:
    - The pharmacy receives the prescription and dispenses the required medicine.
    - The dispensed medicine is tracked in the Pharmacy and Medicine tables.

3. **Pharmacy Inventory Management**:
   - The system monitors the stock levels of medicines.
   - When stock levels are low, the system alerts the pharmacy staff to reorder stock.

Data Flow:

- Prescription → Prescription table
- Pharmacy → Dispense Medicine → Pharmacy and Medicine tables
- Inventory Tracking → Medicine table

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8. Room and Staff Management Workflow

Steps:

1. **Room Assignment**:
   - For admitted patients, the system assigns a room based on availability.
   - The room assignment is tracked in the Room table.
2. **Staff Scheduling**:
   - Hospital staff, including nurses and doctors, have their shifts scheduled and managed by the system.
   - The schedule is stored in the Staff and Department tables.
3. **Monitoring Room and Staff Availability**:
   - The system tracks room availability and staff schedules.
   - Hospital administrators can view available rooms and schedule adjustments for staff.

Data Flow:

- Patient Admission → Room Assignment → Room table
- Staff Scheduling → Staff and Department tables
- Room and Staff Availability → Room, Staff tables

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9. Reporting and Analytics Workflow

Steps:

1. **Patient Reports**:
   - Generate detailed reports on patient history, including medical records, lab results, and prescriptions.
   - Data is retrieved from the MedicalRecord, Prescription, and LabReport tables.
2. **Financial Reports**:
   - The system generates billing and payment reports based on data from the Invoice and Payment tables.
   - Hospital revenue, outstanding payments, and insurance claims are analyzed.
3. **Hospital Performance Reports**:

    - Administrators can monitor hospital performance through metrics like room occupancy, staff efficiency, and patient load.
    - Data is collected from the Room, Staff, and Appointment tables.

Data Flow:

- Patient History Report → MedicalRecord, LabReport, Prescription
- Financial Report → Invoice, Payment
- Performance Report → Room, Staff, Appointment

End-to-End Workflow Example:

1. **Patient Registration** → Login → Schedules an **Appointment**.
2. Doctor accepts appointment (in-person or telemedicine).
3. **Appointment confirmed**. For telemedicine, a video link is generated.
4. The consultation takes place. During the consultation:
    - Doctor views patient **MedicalRecord**.
    - Doctor issues a **Prescription** or requests a **LabReport**.
5. An **Invoice** is generated after the consultation, and the patient proceeds to make a **Payment**.
6. If necessary, the patient collects medicine from the **Pharmacy**.
7. Hospital staff manage **Rooms** and **Staff shifts** for admitted patients.
8. Administrators generate **Reports** for financial, patient, and hospital performance analytics.

This **workflow** ensures a streamlined process from patient admission, appointment management, telemedicine, medical records, billing, pharmacy, room management, and reporting, covering all critical operations in a hospital environment.